October 12–16, 2026

Science Team Weekly

New Bedford High School · Science Department

CLASSROOM INVESTIGATION · AP CHEMISTRY

When Light Becomes Evidence

AP Chemistry students use spectrophotometry, calibration curves, and quantitative reasoning to determine the concentration of an unknown solution.

In Ms. Jayni Gryss's AP Chemistry class, students investigated Beer's Law with a Vernier SpectroVis spectrophotometer and cobalt(II) chloride solutions. The work connected visible light, absorbance, transmittance, concentration, dilution, and graphical modeling.

Students prepared known solutions, identified a wavelength of maximum absorbance, built a calibration curve, and used the resulting linear relationship to determine the concentration of an unknown. A color-based observation became quantitative evidence.

AP Chemistry students use a Vernier SpectroVis instrument, data tools, and laboratory notes during a Beer's Law investigation.

What students are doing: measuring absorbance, modeling a relationship between concentration and light, and using a calibration curve to defend a scientific claim about an unknown sample.

FROM MEASUREMENT TO EXPLANATION

The investigation asks students to explain not only what a graph shows, but why the graph gives evidence about dissolved particles in a solution. That shift—from collecting a measurement to interpreting its meaning—is central to scientific reasoning.

ELD IN SCIENCE · RESOURCE SHARE

Visual Thinking Strategy (VTS)

Thank you to **Ms. Hope Bynoe and Ms. Brittany Jenney** for sharing an instructional resource that builds knowledge and develops student thinking through content-connected visuals.

Visual Thinking Strategy supports students—especially newcomers through proficiency level 1.9—as they observe, speak, listen, and build an explanation before formal writing.

1 · SELECT A VISUAL
Choose a picture, model, graph, specimen, or lab image connected to the lesson.

2 · INVITE NOTICING
Prompt students: "What is going on?" Give time for careful observation.

3 · ASK WHY
Ask students to point to a visible detail that supports an idea or interpretation.

4 · BUILD DISCUSSION
Connect student observations, vocabulary, questions, and scientific thinking.

Language + observation

VTS makes student thinking visible before a teacher asks for a formal claim, calculation, model, or written explanation.

"What do you see that makes you say that?" is a simple question with room for evidence, language, and revision.

This Week in Science

Across AP Chemistry and ELD in Science, students are invited to notice carefully, use evidence, explain their reasoning, and revise their thinking as new information becomes available.

Observe precisely

Students notice a pattern in a solution, image, graph, or sample before deciding what that pattern means.

Use evidence

Measurements, models, and visible details become meaningful when students explain how they support a claim.

Revise thinking

Discussion, feedback, and new information create a reason to strengthen an explanation rather than simply finish a task.

Classroom Spotlight

Students use models, discussion, and evidence to make sense of ecosystem relationships

BIOLOGY · MR. JOSEPH MIKSCH

A Food Web Becomes a System Model

Students traced energy through a marine ecosystem by building a gray-seal food web that made organisms, relationships, and population effects visible.

BUILDING THE WEB

Students in Biology explored ecosystem interactions and dynamics by building a Marine Food Web that included the gray seal, its prey, its predators, and the organisms connected across the ecosystem. Their models used drawings, labels, and arrows to represent producers, consumers, and the direction energy moves.

As groups placed organisms in their models, they moved beyond isolated vocabulary. Students discussed how a food web can show relationships among food, habitat, biodiversity, population dynamics, and carrying capacity.

TESTING A CHANGE

The completed model becomes a tool for reasoning. Students can ask what might happen if the gray-seal population changes, which organisms could be affected first, and what additional evidence would be needed before making a claim.

Why it matters: A useful scientific model does not simply show objects. It helps students explain relationships, test an idea, and revise their thinking when evidence changes.

Ask a student

"Which arrow in your food web would you investigate first if one population changed, and what evidence supports that choice?"

Making the model informative

Students can check whether every arrow has a clear direction, every organism has a scientific role, and every proposed change is connected to a relationship visible in the model. These checks turn a drawing into a tool for explaining ecosystem change.

Students organize organisms and relationships while building a marine food-web model centered on the gray seal.

Vocabulary in use

- Producer and consumer
- Predator and prey
- Energy flow
- Keystone species
- Carrying capacity

Science practice: read, model, discuss, explain, and revise. Each move helps students turn an ecosystem diagram into an evidence-based explanation.

Notice a relationship

Trace one arrow and name the organisms connected by that energy transfer. What does the arrow communicate?

Predict a change

Choose one population increase or decrease. Which relationship in the web could change next?

Support a claim

Return to the model, an ecological idea, and a visible arrow before deciding what the evidence supports.

From model to argument

After building a food web, students can use the model as evidence for a structured explanation: name a change, identify the relationship it affects, explain the likely pathway through the system, and qualify what additional evidence would strengthen the claim.

SENTENCE FRAME

"If the population of ___ changes, then ___ may be affected because the food web shows ___. To be more certain, scientists would need evidence about ___."

Learning Beyond the Shoreline

Cuttyhunk STEAM Academy brings Marine Science inquiry into the science classroom

GUEST LEARNING EXPERIENCE · MICHELLE CARVALHO

Marine Debris as Evidence

Students use materials, fragments, biological clues, and possible pathways to ask what marine debris can—and cannot—reveal.

Students sort and examine marine debris while working through a structured evidence-collection process.

Michelle Carvalho, Program Director of Cuttyhunk STEAM Academy, visited students to introduce a hands-on Marine Science investigation that treats debris as evidence of human activity, environmental processes, and possible pathways into the ocean.

FIELD EVIDENCE

Teams sorted debris by primary material—plastic, metal, glass or ceramic, rubber, fabric, and other—then counted items and calculated the proportion represented by each category. They considered recognizable objects, fragments, and identifying clues before proposing likely original uses or possible routes into the ocean.

OBSERVE BEFORE INFERRING

Using hand lenses, students looked for attached algae, barnacles, abrasions, discoloration, entanglement, and other visible clues. The work emphasized a scientific distinction: an observation is what can be seen; an interpretation is a possible explanation; uncertainty names what the evidence cannot yet establish.

From observation to action

Marine Science connects environmental science, ecology, materials science, and data analysis while giving students a real-world reason to observe carefully and ask better questions.

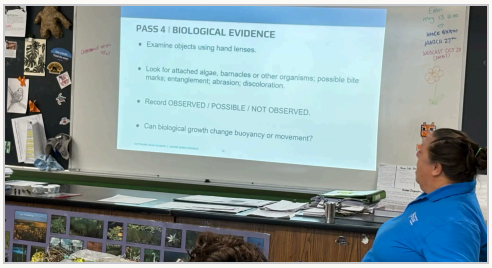
Marine Science connection

Materials can move through coastal systems, weather into fragments, collect organisms, and interact with wildlife. Students use the debris sample to connect local observations to larger questions about human choices, ocean transport, and the health of marine ecosystems.

TRACE
What path might bring this object from a source to the ocean?

COMPARE
Which physical or biological clues support one possible pathway?

QUESTION
What important detail is still missing before drawing a conclusion?



Michelle Carvalho introduces the Marine Science investigation.

Evidence cycle

01 **Materials & counts**
Sort, tally, and quantify what is present.

02 **Objects & sources**
Compare fragments with possible original uses and pathways.

03 **Biological evidence**
Record what is observed, possible, or not observed.

Guiding question: What can a collection of marine debris reveal about sources, transport, weathering, and interactions with marine life?

Observed

Record a material, mark, organism, count, or other visible feature without adding an explanation.

Possible

Offer a careful interpretation that is consistent with the evidence while recognizing other explanations may exist.

Not yet known

Name the additional observation, data source, or test that would be needed before reaching a stronger conclusion.

Science Corner

Everyday chemistry, microbial metabolism, food preservation, and scientific caution

CHEMISTRY IN EVERYDAY LIFE

The Power of Vinegar: A Sour Solution with a Long Scientific Story

Vinegar links fermentation, acetic acid, food preservation, and a long history of antimicrobial investigation—while reminding scientists to distinguish a useful effect from an overclaim.

WHAT TURNS JUICE SOUR?

Vinegar is mostly water plus **acetic acid**, the molecule responsible for its sour taste, sharp aroma, and many practical uses. Common culinary vinegars generally contain about 4–8% acetic acid [4].

Its production is a two-stage microbial process. First, yeast convert sugars from grapes, apples, rice, or grains into ethanol. Then acetic-acid bacteria use oxygen to convert ethanol into acetic acid [1].

WHY ACIDITY MATTERS

Acetic acid lowers pH and can make conditions less favorable for many spoilage microbes. This is one reason vinegar has been used for centuries in pickling and food preservation. In some research settings, acetic acid may also challenge microbial metabolism and biofilms—organized microbial communities attached to surfaces [2].

Try · observe · explain

01 **Try:** Compare a fresh vegetable and a safely prepared quick-pickled sample.

02 **Observe:** Notice changes in texture, odor, and the acidic environment.

03 **Explain:** Connect pH to microbial conditions and food preservation.

What acidity changes

Acetic acid changes the environment surrounding a cell. The lower pH can make it harder for many microbes to maintain the internal conditions needed for growth, which helps explain why acidity is useful in carefully prepared food preservation [1][2].

Vinegar connects fermentation chemistry, food preservation, and a long history of scientific curiosity.

A LONG MEDICAL HISTORY—WITH AN IMPORTANT DISTINCTION

Historical records and later clinical practice describe dilute acetic-acid solutions as topical antiseptics. During World War I, clinicians reported use of a 1% acetic-acid solution against *Pseudomonas* wound infections. Modern research continues to study clinical-grade acetic acid for selected wound-associated bacteria and biofilms [3].

Important modern distinction: Vinegar is **not a sterilizing agent** and should not be treated as a do-it-yourself wound treatment. Household vinegar may kill some microbes, but it does not eliminate them all and is not a substitute for approved disinfectants or professional wound care [4].

SELECTED REFERENCES

[1] Chen, G.-L., et al. (2023). *Vinegar: A potential source of healthy and functional food with special reference to sugarcane vinegar.* Frontiers in Nutrition, 10.

[2] Harrison, F., et al. (2023). *Exploring the antibacterial and antibiofilm activity of acetic acid and vinegar against wound-associated bacteria.* Microbiology, 169.

[3] Agrawal, K. S., et al. (2017). *Acetic acid dressings: Finding the Holy Grail for infected wound management.* Indian Journal of Plastic Surgery, 50(3), 317–321.

[4] Harvard T.H. Chan School of Public Health. (2019). *Vinegar.* The Nutrition Source.

Science in everyday life

Vinegar begins with microbial metabolism, becomes an acidic chemical environment, and connects to food preservation and medical history. Its story also shows why scientists distinguish a real antimicrobial effect from the much stronger claim of sterilization.

Classroom connection

Vinegar is a useful case study for the relationship among microbial metabolism, pH, chemical environment, food preservation, and the careful limits of a scientific claim.

October 12–16, 2026

Science Team Recognition

Collaboration, preparation, access, and shared commitment across the department

A DEPARTMENT THAT LEARNS TOGETHER

The Work Behind Strong Science Learning

This week's recognitions celebrate colleagues who strengthen classroom learning through coaching, curriculum work, resources, co-planning, field experiences, and a shared commitment to students.

Friday coaching exchange

Mr. Kevin Flynn and Mr. Nathaniel Walker—thank you for exchanging ideas and sharing teaching experiences during Friday coaching sessions. Professional reflection helps colleagues learn from one another and grow as educators.

Resources, materials & lesson coherence

Mr. Tyler Gilbert—thank you for distributing resources, including notebooks, to OpenSciEd teachers and for helping make the department a better working environment. **Mr. Brian McCauley and Mr. Tyler Gilbert**—thank you for checking standards alignment and reminding colleagues to connect lessons to future learning.

Field learning & physics curriculum

Ms. Patricia Gifford—thank you for planning AP Environmental Science field trips and extending real-world learning experiences to Marine Science students. **Mr. Nicholas LeBlanc**—thank you for piloting OpenSciEd Physics and sharing those experiences in PLCs.

Co-teaching partnership

Mr. Jacob Ramos, Ms. Cykka Lovett, Mr. Matthew Golden, Mr. Richard Dwinell, and all co-teachers—thank you for collaborative planning and idea exchange that anticipates accommodations and supports full student participation.

Generosity with shared materials

Ms. Cassidy Barrett—thank you for generously allowing department materials to be temporarily stored in your room before pickup. Your flexibility helped keep resources organized and available for colleagues.

How collaboration reaches students

When teachers plan together, students benefit from materials that are ready, lessons that connect across a unit, supports that anticipate needs, and feedback that helps them take a meaningful next step. Department collaboration is ultimately an investment in more coherent and accessible science learning.

Prepare resources before instruction begins.

Connect daily lessons to the next learning target.

Respond together to evidence of student thinking.

Biology curriculum leadership

Ms. Grace Stevenson—thank you for leading the Biology OpenSciEd group in adapting units for standards alignment and curriculum development.

Newcomer teaching expertise

Ms. Seannine Tveit and Ms. Lisa Richard—thank you for sharing your expertise in teaching newcomers and strengthening Biology instruction for all learners. Your language-aware practices help colleagues plan lessons that are both rigorous and welcoming.

Biology team cohesion

Thank you to all Biology teachers for the commitment to collaborative OpenSciEd implementation and a cohesive culture that supports students and colleagues.

Chemistry team collaboration

Ms. Samantha Roche, Ms. Jayni Gryss, and Ms. Margaret Henning—thank you for demonstrating the power of PLC collaboration through pacing, materials sharing, and standards alignment.

What our department makes possible

When colleagues share ideas, strengthen access, prepare materials, and connect lessons to meaningful experiences, students encounter a science department ready to support both curiosity and high expectations.

Collaboration

Coaching, PLC discussion, and shared planning make individual expertise available to the full department.

- Share a strategy, resource, or lesson insight.
- Ask a colleague for feedback before the next lesson.

Coherence

Materials preparation, pacing, and standards alignment help students experience connected learning across courses.

- Connect today's task to the next learning target.
- Protect time for high-quality materials and planning.

Access

Co-teaching partnerships, newcomer expertise, field learning, and thoughtful supports widen meaningful participation in science.

- Anticipate a support before students need it.
- Maintain high expectations through multiple pathways.

The Science Department signature

Students benefit when a department treats curriculum, materials, collaboration, accessibility, and real-world learning as shared responsibilities. Thank you to the many colleagues whose daily contributions make that work possible.

Shared Practices, Stronger Learning

Practical routines that make science thinking visible, accessible, and actionable

A PRACTICAL FINISH TO THE WEEK

Make the Next Step in Learning Clear

Across AP Chemistry, Biology, Marine Science, and ELD in Science, students benefit when lessons make observation, evidence, discussion, and revision visible.

NBHS ELD IN SCIENCE

Instructional Strategies for English Learners

Thank you to **Ms. Hope Bynoe and Ms. Brittany Jenney** for sharing Visual Thinking Strategy (VTS), a resource that uses content-connected visuals to build knowledge, language, and student thinking before formal writing.

01 Launch learning with a visual

Begin with a picture, model, graph, specimen, or investigation image that is directly connected to the science learning.

Use it when: *inviting students to ask, "What is going on?" and "What do you see that makes you say that?"*

- Give quiet notice time before discussion.
- Record student words, then connect them to science vocabulary.
- Return to the visual when students write or revise.

02 Plan for evidence, not just completion

Identify where student thinking will become visible: a model, data table, discussion, explanation, calculation, or product.

Use it when: *deciding what student evidence will guide the next instructional move.*

- Name the success criterion before the task begins.
- Anticipate one likely misconception or incomplete idea.
- Select a student product to revisit with the team.

03 Give feedback that prompts action

Feedback is most useful when it connects to a goal, names one manageable next move, and leaves the cognitive work with the learner.

Use it when: *students can revise a claim, add evidence, clarify a model, compare results, or strengthen an explanation.*

- Connect the comment to a goal or success criterion.
- Ask for one visible revision—not a general improvement.
- Protect time for students to use the feedback.

04 Create multiple pathways into rigor

Rigorous science becomes more accessible when students encounter an idea through language, visuals, models, investigation, discussion, and writing.

Use it when: *planning vocabulary, representations, partner talk, and scaffolds without lowering scientific demand.*

- Pair a visual, model, or data display with structured talk.
- Offer a sentence frame that supports a scientific explanation.
- Keep the same learning goal while varying the pathway.

Across the team: AP Chemistry students used light data to model concentration; Biology students used a food web to model ecosystem relationships; Cuttyhunk STEAM Academy visitors used debris clues to model environmental pathways. Different subjects, one shared practice: use evidence to build and revise explanations.

Looking ahead

Every strong learning experience gives students a way to notice, discuss, explain, and improve. The shared work is to design those opportunities with clarity, access, and high expectations.

Data & calibration

AP Chemistry students compare known concentrations, identify a pattern, and use a calibration curve—rather than a guess—to justify the concentration of an unknown solution.

Student move: explain why the line of best fit is useful evidence.

Models & systems

Biology students use a food web to predict a change and locate the arrows, relationships, and evidence needed to support an ecosystem claim.

Student move: point to an arrow and explain the pathway of impact.

Observation & uncertainty

Marine Science students separate what is observed from what is possible and what remains uncertain, keeping explanations careful and credible.

Student move: name one additional clue needed for a stronger conclusion.

Next-step question

Where will students make their thinking visible? Name the moment, product, or conversation that will reveal their reasoning.

Plan: before the lesson, during the task, or at the revision point?

Evidence look-for

What student example, pattern, or scientific idea will show whether the planned support helped?

Collect: one work sample, discussion note, model, or data response.

THE DEPARTMENT LEARNING CYCLE

Plan together · Make thinking visible · Respond with feedback · Adjust together

Plan with clarity, **notice** evidence of student thinking, **respond** with one usable next step, and **adjust** the next learning experience together.

Plan · materials & supports | **Notice** · student evidence | **Respond** · actionable feedback | **Adjust** · the next lesson